Commodore 64 & Family

BASIC 1.0–7.0 | PET · VIC-20 · C64 · C128 · C16 · PLUS/4 · C64 ULTIMATE

Conventions: device **8**=first disk, **1**=tape. C= is the Commodore key. Cards tag the BASIC version where features differ: **2.0** VIC/64, **3.5** C16/Plus4, **7.0** C128.

PROGRAM CONTROL — direct mode

RUN	Start program; RUN 100 from a line
LIST	Show program; LIST 10-50
NEW	Erase program in memory
CLR	Clear variables, keep code
CONT	Resume after STOP / RUN-STOP
STOP / END	Halt program
SYS addr	Call machine-code at addr
WAIT a,m,x	Pause until mem bit set

BASIC 3.5/7.0 add AUTO, RENUMBER, DELETE, HELP, TRON/TROFF.

BASIC STATEMENTS

PRINT ?	; joins, , tabs columns
INPUT GET	Read line / single key
LET	Assign (optional)
REM	Comment
GOTO GOSUB	Jump / call; RETURN
ON x GOTO	/ GOSUB a,b,c
FOR..NEXT	FOR I=1 TO 9 STEP 2
IF..THEN	IF A>5 THEN 200
DATA READ	RESTORE rewinds DATA
DIM	DIM A(20),N$(5)
DEF FN	DEF FNS(X)=X*X
POKE a,v	Write byte to memory

Multi-statement line: separate with : · 3.5/7.0 add IF..ELSE, DO/LOOP.

FUNCTIONS

ABS SGN INT	magnitude / sign / floor
SQR EXP LOG	root, e^x, ln
SIN COS TAN ATN	radians
RND(x)	RND(0) timer-seeded
PEEK(a)	read a byte
FRE(0)	free BASIC bytes
LEN POS	string len / cursor col
LEFT$ RIGHT$ MID$	MID$(A$,3,2)
STR$ VAL	num↔string
CHR$ ASC	code↔char (PETSCII)
TAB SPC	in PRINT only
USR(x)	call ML at vector $0311

DISK & TAPE I/O — device 8 = disk

LOAD "name",8	Load program from disk
LOAD "name",8,1	Load to its saved address
LOAD "$",8 : LIST	Show disk directory
SAVE "name",8	Save program
VERIFY "name",8	Compare prog to disk
LOAD / SAVE	Tape (device 1, no ,8)
OPEN 15,8,15	Open command channel
OPEN 2,8,2,"f,s,r"	Seq file read
PRINT# / INPUT#	Write / read open file
CLOSE n	Close logical file n

LOAD"*",8,1 loads first program on disk to its address.

DISK DOS COMMANDS — via ch.15

N0:name,id	Format (NEW) disk
S0:name	Scratch (delete) file
R0:new=old	Rename file
C0:new=old	Copy file
V0	Validate (clean BAM)
I0	Initialize drive

Send via OPEN15,8,15:PRINT#15,"S0:FILE". BASIC 3.5/7.0 & PET/CBM 4.0 keywords: DLOAD DSAVE SCRATCH DIRECTORY HEADER RENAME COPY COLLECT.

SCREEN & CURSOR CODES — PRINT CHR$()

CHR$(147)	Clear screen (SHIFT+CLR)
CHR$(19)	HOME cursor
CHR$(17) (145)	Cursor down / up
CHR$(29) (157)	Cursor right / left
CHR$(18) (146)	Reverse on / off
CHR$(14) (142)	Lower-case / upper-graphics
CHR$(13)	RETURN
CHR$(34)	Quote " (toggles quote mode)

Inside strings these show as reverse-video characters — type them with the matching key.

PETSCII TEXT COLORS — CHR$ / CTRL+#

144 Black	5 White
28 Red	159 Cyan
156 Purple	30 Green
31 Blue	158 Yellow
129 Orange	149 Brown
150 Lt Red	151 Dk Grey
152 Grey	153 Lt Green
154 Lt Blue	155 Lt Grey

Hold CTRL+1–8 or C=+1–8 to set the text colour as you type.

C64 MEMORY & POKES — $D000 VIC

53280	Border colour (0–15)
53281	Background colour
646	Current character colour
1024–2023	Screen RAM (40×25)
55296+ ($D800)	Colour RAM
53272	Screen/char-set pointer
197	Key currently pressed
198 / 631	Keyboard buffer count / data
56320 / 56321	Joystick port 2 / 1
SYS 64738	Reset the C64

VIC-20 differs — see Quick-Facts. POKE colour into 646 before PRINT.

SID SOUND (C64) — $D400=54272

54296	Volume 0–15: POKE 54296,15
54272 / 54273	Voice-1 freq lo / hi
54277	Attack / Decay
54278	Sustain / Release
54276	Waveform + gate

Set ADSR + freq, then POKE 54276,17 (triangle+gate) to play; 33=saw, 65=pulse, 129=noise; even value = note off.

SPRITES (C64) — $D000 VIC

53269	Enable: bit per sprite 0–7
2040–2047	Sprite data pointers (×64)
53248 / 53249	Sprite-0 X / Y position
53264	X bit-8 (for X>255)
53287–53294	Sprite 0–7 colours
53271 / 53277	Expand Y / X
53278 / 53279	Sprite-sprite / -data hit

Data lives at pointer×64; each sprite is 24×21 pixels (63 bytes).

BASIC 3.5 / 7.0 GRAPHICS — C16 +4 128

GRAPHIC m[,c]	0 text,1 hi-res,2 split,3 multi
COLOR s,c[,l]	Set source colour
DRAW c,x,y TO x,y	Line
CIRCLE c,x,y,rx,ry	Ellipse
BOX c,x1,y1,x2,y2	Rectangle
PAINT c,x,y	Flood fill
CHAR c,r,col,"t"	Text at row/col
LOCATE x,y	Move pixel cursor
SSHAPE/GSHAPE	Save / stamp area to A$
SCNCLR	Clear graphics screen

SOUND & STRUCTURED — 3.5 / 7.0

SOUND v,f,d	Tone (C16/+4): voice,freq,dur
VOL n	Volume 0–8
PLAY "..."	128 music string + TEMPO
ENVELOPE / FILTER	128 SID shaping
DO..LOOP	WHILE/UNTIL/EXIT
IF..THEN..ELSE	One-line branch
GETKEY v$	Wait for a key
TRAP n / RESUME	Error handling

128-only: SPRITE MOVSPR SPRDEF COLLISION; GO64 → C64 mode; FAST/SLOW 2MHz.

MACHINE QUICK-FACTS

PET / CBM	BASIC 1.0–4.0 · mono · IEEE-488 · scr 32768 · no sprites/SID
VIC-20	BASIC 2.0 · scr 7680 · col 38400 · border+bg POKE 36879 · ~3.5K
C64	BASIC 2.0 · scr 1024 · col 55296 · SYS 64738 reset · 38K
C16	BASIC 3.5 · TED video+sound · 12K free
Plus/4	BASIC 3.5 · built-in 3-Plus-1 apps · 60K free
C128	BASIC 7.0 · 40/80-col · GO64 · FAST · 122K
C64 Ultimate	2025 FPGA C64 · Ultimate-II fw · WiFi · HDMI · see panel

EDITING & SPECIAL KEYS

RUN/STOP	Break running program
RUN/STOP+RESTORE	Warm reset to BASIC
SHIFT+CLR/HOME	Clear screen
C= + SHIFT	Toggle upper / lower case
INST/DEL	Insert / delete char
CTRL (hold)	Slow list / pick colour
f1–f8	Function keys = CHR$(133)–(140)
Quote mode	Edit codes literally after a "

C64 ULTIMATE — 2025 · Ultimate fw

C= + RESTORE	Toggle the Ultimate menu overlay
Power button (tap)	Also opens menu; C64 keeps running
Cursors / W A S D	Navigate; RIGHT enters, RUN/STOP backs out
F1 (in menu)	Power & Reset → Reset or Power OFF
F2 (in menu)	Advanced settings
C= + J (in menu)	Swap joystick ports 1 ↔ 2
C= + N / C= + P	Hold at power-on → NTSC / PAL
RUN/STOP+RESTORE	Warm reset, memory kept

Runs the Ultimate-II firmware (same menu as the Ultimate 64 / 1541 Ultimate). Hold the power button ~4 s for a normal shutdown.